Fiscal Year:	FY 2008	**Task Last Updated:**	FY 07/30/2009
PI Name:	Swain, Greg Ph.D.		
Project Title:	Advanced diamond electrodes for electrochemical based monitoring of spacecraft water		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	517-355-9715 229
Organization Name:	Michigan State University		
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PI Address 2:	320 Chemistry Building		
PI Web Page:	http://www.chemistry.msu.edu		
City:	East Lansing	**State:**	MI
Zip Code:	48824	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	08/26/2004	**End Date:**	08/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 08/31/2008 per PI (9/2007)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schultz, John (Wyle Laboratory) Rutz, Jeffrey (Wyle Laboratory)		
Grant/Contract No.:	NNJ04HI10G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The research is focused on developing advanced water quality monitoring methods for the potable and technical water supplies onboard the International Space Station. The methods being developed could also have broad application on other long-duration space flight and habitation projects. Electrochemical methods are being developed for water quality monitoring with two contaminant types targeted: (i) heavy metal ions including Ag(I), Cu(II), Cd(II), Pb(II) and Zn(II) and (ii) iodine and iodide as well as phenol and iodinated phenols. The key enabling material is the boron-doped diamond electrode, which provides a sensitive, reproducible and stable response for these analytes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The method developed was specifically designed to impact water quality monitoring onboard ISS, but the technology could also provide sensitive, reproducible and stable detection of these, and other contaminants, in surface and ground water.
Task Progress:	We completed studies of the electrode and method for a series of laboratory-prepared standard solutions. We also redesidned a small flow cell that could be used in microgravity environments for water quality assessment. We devised and tested a multiple step analysis protocol that involved sampling the water from the galley, mixing with a prepackaged supporting electrolyte (in a sealed syringe), adding the solution to the cell, making the anodic stripping voltammetric measurements of the metal ion content and analysing the data. Our focus was on Ag(I). We also applyed the method and cell to the analysis of several simulant water samples provided to us by Wyle Laboratories.
Bibliography Type:	Description: (Last Updated: 08/03/2009)
Articles in Peer-reviewed Journals	Sonthalia P, McGaw E, Show Y, Swain GM. "Metal ion analysis in contaminated water samples using anodic stripping voltammetry and a nanocrystalline diamond thin-film electrode." Anal Chim Acta 2004 Sep 20;522(1):35-44. http://dx.doi.org/10.1016/j.aca.2004.06.071 , Sep-2004
Articles in Peer-reviewed Journals	McGaw EA, Swain GM. "A comparison of boron-doped diamond thin-film and Hg-coated glassy carbon electrodes for anodic stripping voltammetric determination of heavy metal ions in aqueous media." Anal Chim Acta. 2006 Aug 11;575(2):180-9. PubMed PMID: 17723589 , Aug-2006
Articles in Peer-reviewed Journals	Song Y, Swain GM. "Total inorganic arsenic detection in real water samples using anodic stripping voltammetry and a gold-coated diamond thin-film electrode." Anal Chim Acta. 2007 Jun 12;593(1):7-12. PubMed PMID: 17531818 , Jun-2007
Articles in Peer-reviewed Journals	Song Y, Swain GM. "Development of a method for total inorganic arsenic analysis using anodic stripping voltammetry and a Au-coated, diamond thin-film electrode." Anal Chem. 2007 Mar 15;79(6):2412-20. PubMed PMID: 17302386 , Mar-2007